Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Generative AI–Based Multimodal Digital Therapeutics for Obesity and Metabolic Disease Management through Digital Phenotype Analysis			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Electrical Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Information & Systems	Other Medical Information & Systems	Measuring Instruments	Measurement Sensors and Components
Concept of Proposed Technology	The proposed technology is a **multimodal digital therapeutic (DTx) platform** designed to support sustainable management of obesity and metabolic diseases through **AI-driven behavioral intervention and digital phenotype analysis**. The platform integrates: • **Portable body composition and biosignal measurement** technologies (e.g., BIA and PPG) for continuous physiological monitoring • **Small language model (sLLM)–based generative AI** enabling real-time, conversational lifestyle coaching and adherence monitoring • **Deep learning–based risk prediction models** that classify obesity endotypes and estimate risks of metabolic complications • A **cloud-based clinician dashboard** that allows healthcare professionals to monitor patient adherence, behavioral patterns, and risk trajectories			

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	By combining objective physiological data with AI-driven behavioral therapy grounded in cognitive behavioral therapy (CBT) principles, the system provides a closed-loop digital intervention designed to complement pharmacological treatments and support long-term lifestyle modification.
Rationale for Joint R&D Project	Pharmacological treatments for obesity, including GLP-1–based therapies, demonstrate strong short-term efficacy but are associated with **high cost, limited accessibility, and significant weight rebound after discontinuation**. These limitations highlight the need for scalable, cost-effective digital interventions that promote sustained behavioral change and preserve treatment benefits. Effective development of AI-based digital therapeutics requires: • Large-scale, **multi-ethnic cohort data** to ensure model robustness and global applicability • Independent **clinical validation in international settings**, particularly for regulatory acceptance in overseas markets • Optimization of AI-driven interventions to reflect cultural and behavioral differences across populations International joint R&D enables access to diverse cohort datasets and advanced clinical research infrastructure, supporting global optimization of obesity endotype classification models and cross-validation of clinical effectiveness. This collaborative framework is essential to establish international credibility and regulatory readiness for global deployment.
Key Words	1. Obesity Management 2. Generative AI 3. Korean/Global Obesity Endotype 4. Digital Therapeutics (DTx) 5. Metabolic Disease Risk

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (8)
R&D Objectives	Overall Objective To develop and clinically validate a **generative AI–based multimodal digital therapeutic platform** for obesity and metabolic disease management and advance the technology from **TRL 4 to TRL 8**, enabling readiness for domestic and international regulatory approval. Specific Objectives • Develop AI models that classify obesity endotypes and predict risks of metabolic complications • Implement a CBT-based multimodal digital intervention using generative AI conversational monitoring • Integrate biosignal data, AI models, and clinician dashboards into a unified digital therapeutic platform			

	• Verify clinical effectiveness through domestic confirmatory trials and international cross-validation • Establish regulatory-ready evidence supporting medical device and digital therapeutic submissions
Target Specifications	Key performance targets include: • **Metabolic Risk Prediction Accuracy**: AUC ≥ 0.77 • **Lifelog and Biosignal Data Extraction Accuracy**: ≥ 85% • **Clinical Trial Scale**: ≥ 20 total subjects across exploratory and confirmatory studies • **International Cross-Validation**: ≥ 2 completed cross-validation sessions using overseas clinical datasets
R&D Plan	The R&D program is structured over **36 months** in three sequential phases. Phase 1: AI Modeling and Core System Architecture (Months 1–12) • Standardization and preprocessing of long-term cohort data to define obesity endotypes and metabolic health categories • Development of deep neural network models predicting metabolic disease risks, targeting AUC ≥ 0.77 • Design of sLLM-based generative AI conversational engines for monitoring nutrition, physical activity, and emotional status • Initiation of multi-ethnic data exchange with overseas research partners to support global model optimization Phase 2: System Integration and Exploratory Clinical Trials (Months 13–24) • Integration of portable biosignal sensors with cloud-based clinician dashboards • Deployment of the unified digital therapeutic platform • Execution of exploratory clinical trials to validate digital phenotypes and closed-loop intervention mechanisms • Software validation and reliability testing in accordance with ISO/IEC 25010 standards • Usability evaluation and cultural adaptation for overseas clinical environments Phase 3: Confirmatory Trials and Global Cross-Validation (Months 25–36) • Conduct of large-scale domestic confirmatory clinical trials to evaluate improvements in BMI and metabolic indicators • Execution of international pilot studies to verify model generalizability across diverse ethnic groups • Preparation of regulatory documentation for domestic authorities and alignment with overseas regulatory pathways • Development of commercialization strategies for global markets

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will focus on **global clinical validation and regulatory support**, including: ▪ Utilization of overseas cohort data to optimize AI-based endotype classification ▪ Leadership of international cross-validation studies and usability assessments ▪ Provision of clinical protocols and datasets supporting overseas regulatory submissions ▪ Contribution to interpretation of clinical outcomes and global deployment strategies
Expected R&R of Korean SME Partner	The Korean SME partner will serve as the **technology development and project lead**, including: ▪ Development of portable biosignal measurement devices and sLLM-based digital therapeutic platforms ▪ Deployment of AI models for metabolic risk prediction and CBT-based intervention ▪ Management of overall R&D execution, domestic clinical trials, and preparation of regulatory submissions ▪ Coordination of data integration, system validation, and commercialization planning

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technological Impact • Establishment of a **first-in-class multimodal digital therapeutic platform** integrating hardware-based biosignal measurement and generative AI–driven behavioral therapy • Establishing a new standard for next-generation digital therapeutics through integrated measurement, analysis, and behavioral intervention Clinical Impact • Improved long-term adherence and reduced weight rebound in obesity patients, particularly when used alongside pharmacological treatments • Enhanced early detection and prevention of metabolic complications Economic and Industrial Impact • Strengthened competitiveness in the global digital therapeutics market • Expansion of export opportunities for Korean digital health technologies Societal Impact • Reduction of socioeconomic burdens associated with obesity and metabolic diseases • Improved accessibility to sustainable, AI-driven lifestyle intervention solutions

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6. References

(1) WHO Report on Obesity: The World Health Organization (WHO) classifies obesity as a chronic disease requiring long-term medical intervention and 21st-century "new epidemic" status

(2) Market Trends (2022): Global Digital Therapeutics Market (2020-2027) reports a 23.3% CAGR for the obesity segment, the highest growth rate among chronic diseases

(3) Clinical Precedence: Research by Seoul National University College of Medicine (2020) demonstrated the efficacy of lifestyle-correction digital therapeutics for weight management.

(4) Pharmacological Synergy: Recent studies on GLP-1 agonists (Semaglutide/Tirzepatide) highlight a significant weight rebound effect upon treatment cessation, necessitating digital behavioral intervention.

(5) Technical Reliability: Validation studies at Ewha Womans University Seoul Hospital confirmed that Onesoftdigm's portable BIA technology shows no significant difference compared to clinical-grade InBody 720 equipment.